House Price Prediction using Machine Learning

This project focuses on predicting house prices using Machine Learning techniques. The Boston Housing dataset was used for analysis, preprocessing, visualization, model building, and evaluation. The project demonstrates how data analytics and machine learning can help estimate property prices based on multiple housing features.

Project Objectives

• Understand the structure of the Boston Housing dataset.
• Perform data preprocessing and exploratory data analysis (EDA).
• Identify relationships between features and house prices.
• Train a machine learning regression model.
• Evaluate model performance using regression metrics.

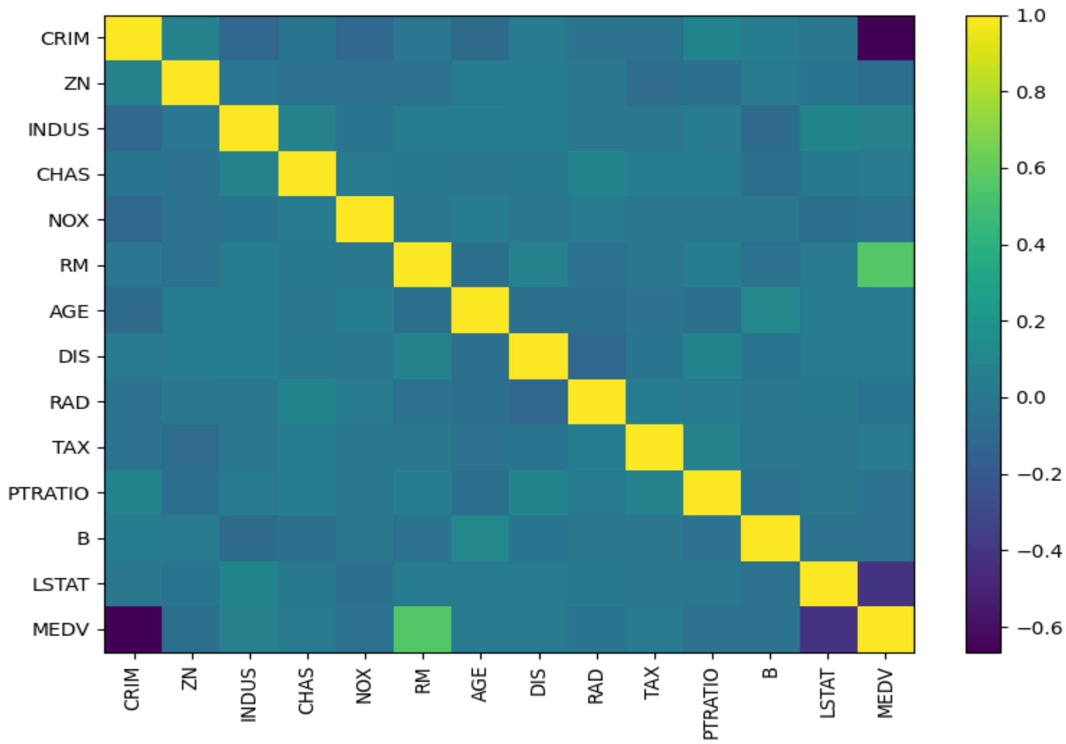
Dataset Overview

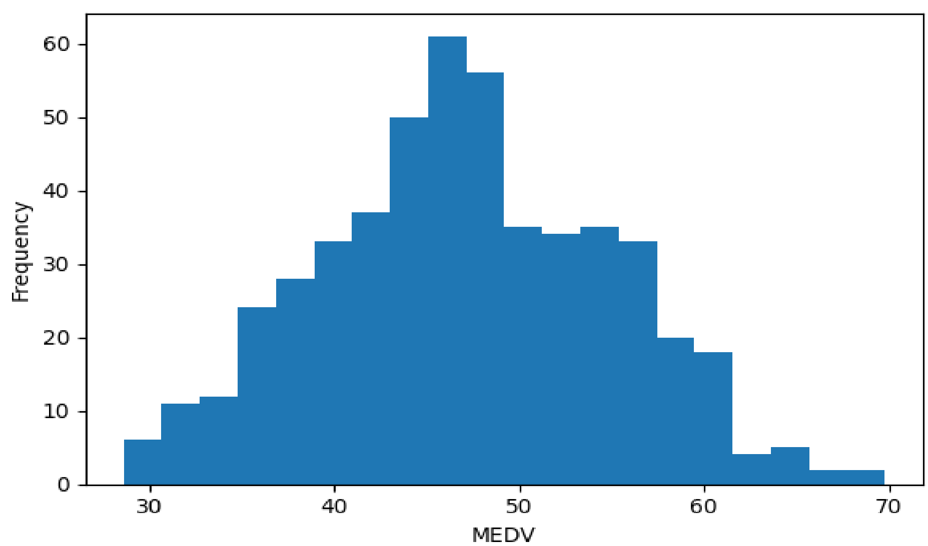
The dataset contains **506** rows and **14** columns. The target variable is **MEDV**, representing the house price value.

CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
33.715	91.09	28.35	0.0	0.771	6.04	79.52	3.24	11.0	545.0	12.01	129.84	28.82	42.27
85.565	82.25	1.66	1.0	0.835	6.8	85.52	1.74	12.0	486.0	23.74	269.1	22.77	38.84
65.882	94.98	21.31	0.0	0.873	3.43	24.96	10.65	12.0	748.0	12.32	134.91	8.87	37.71
53.883	72.57	27.79	0.0	0.772	4.53	96.1	5.28	7.0	291.0	14.72	71.84	4.59	45.68
14.05	61.34	5.83	0.0	0.489	5.17	20.5	6.96	13.0	466.0	17.41	112.54	3.38	54.31

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of variables and identify relationships among features. Correlation analysis helps determine which variables strongly influence house prices.





Machine Learning Model

The project uses the XGBoost Regressor algorithm for prediction. The dataset was split into training and testing sets using train_test_split. The model learns patterns from the training data and predicts house prices on unseen data.

Project Workflow

1. Import libraries and dataset.
2. Perform data cleaning and preprocessing.
3. Analyze dataset statistics and correlations.
4. Split dataset into training and testing sets.
5. Train XGBoost Regressor model.
6. Predict house prices.
7. Evaluate model using R^2 Score and Mean Absolute Error.

Advantages of the Project

• Helps estimate property prices accurately.
• Demonstrates real-world machine learning workflow.
• Useful for real estate analytics and business decision-making.
• Enhances understanding of regression algorithms.

Conclusion

The House Price Prediction project successfully demonstrates the use of machine learning for regression analysis. Using the Boston Housing dataset and XGBoost Regressor, the model can predict housing prices effectively. The project provides practical experience in data preprocessing, visualization, machine learning model development, and evaluation.

Future Scope

Future improvements may include using larger real-world datasets, deploying the model as a web application, integrating advanced algorithms, and optimizing hyperparameters for better prediction accuracy.